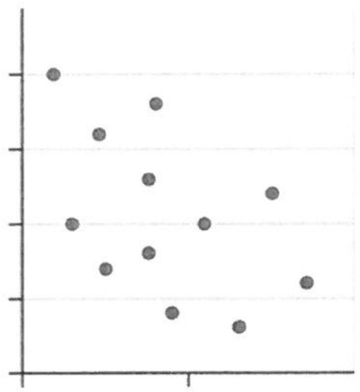
<table>
<tr><td>Name:

Teacher :</td><td>____________________

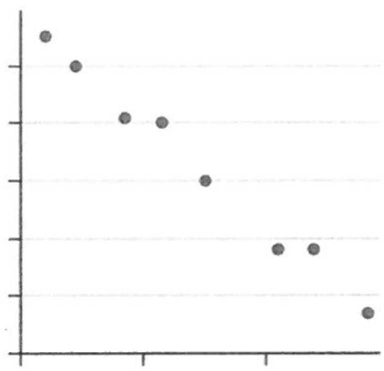
____________________</td><td>Date:________</td></tr>
<tr><td>Baldivis
Secondary College</td><td colspan="2">Year 10 <s>General</s> A
Mini Test 5
Topic – Bivariate Data
SCORE: / 38

<u>Full working out MUST be shown to get full marks for each question.</u></td></tr>
<tr><td>Total Time:</td><td>40 minutes</td><td rowspan="2">20 mnutes calc free , 20 minutes calc allowed.</td></tr>
<tr><td>Weighting:</td><td>5%</td></tr>
<tr><td>Equipment:</td><td colspan="2">To be provided by the student: Pen, pencil, ruler, scientific calculator, 1 single sided page of A4 notes</td></tr>
</table>

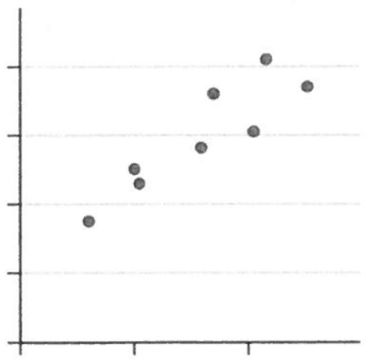
Q1) Describe the relationship for the following scatter graphs. **[6 marks]**



a. Very weak negative.



b. linear, negative, strong.



c. linear, positive Moderate.

Q2) Using your knowledge of Bivariate data **[1 + 2 = 3 marks]**

a. Define the term "Causal Relationship"

One variable effects the other

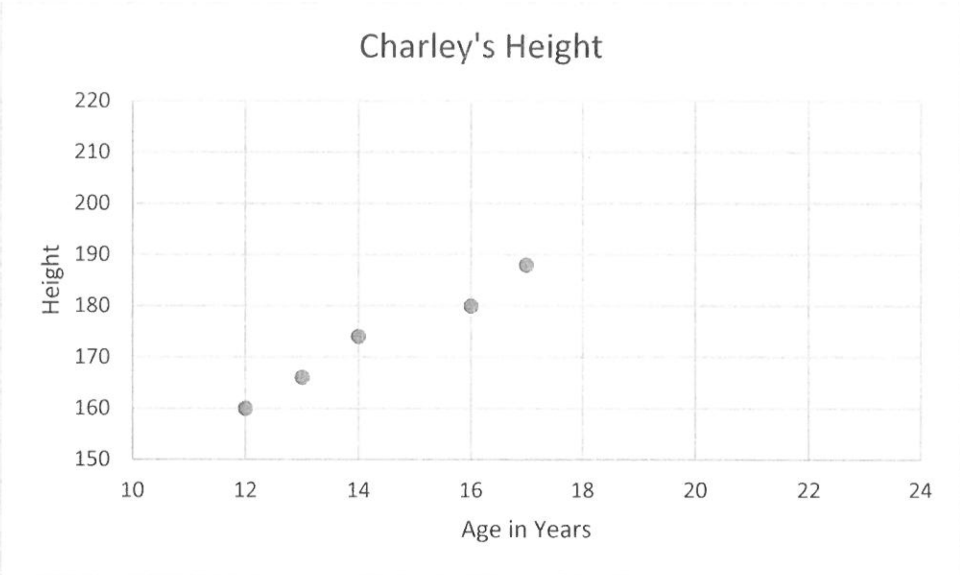
b. Provide an example of a causal relationship and explain how it is causal.

~~Ka~~ Example. / Explain

Q3) Charley recorded her height from the time she was 12 to 17.

[2 + 1 + 1 +1 = 5 Marks]

Age (years)	**12**	**13**	**14**	**16**	**17**
Height (cm)	**160**	**166**	**174**	**180**	**188**



a. Draw in a line of best fit, make a prediction about how tall she will be at 20 years old?

✓ Using line ✓

b. Do you think that your prediction is accurate?

No ✓

c. What do you think would be a reasonable height for charley at 20 years old? Why?

Reasonable answer ✓

d. How tall was Charley at 15 years?

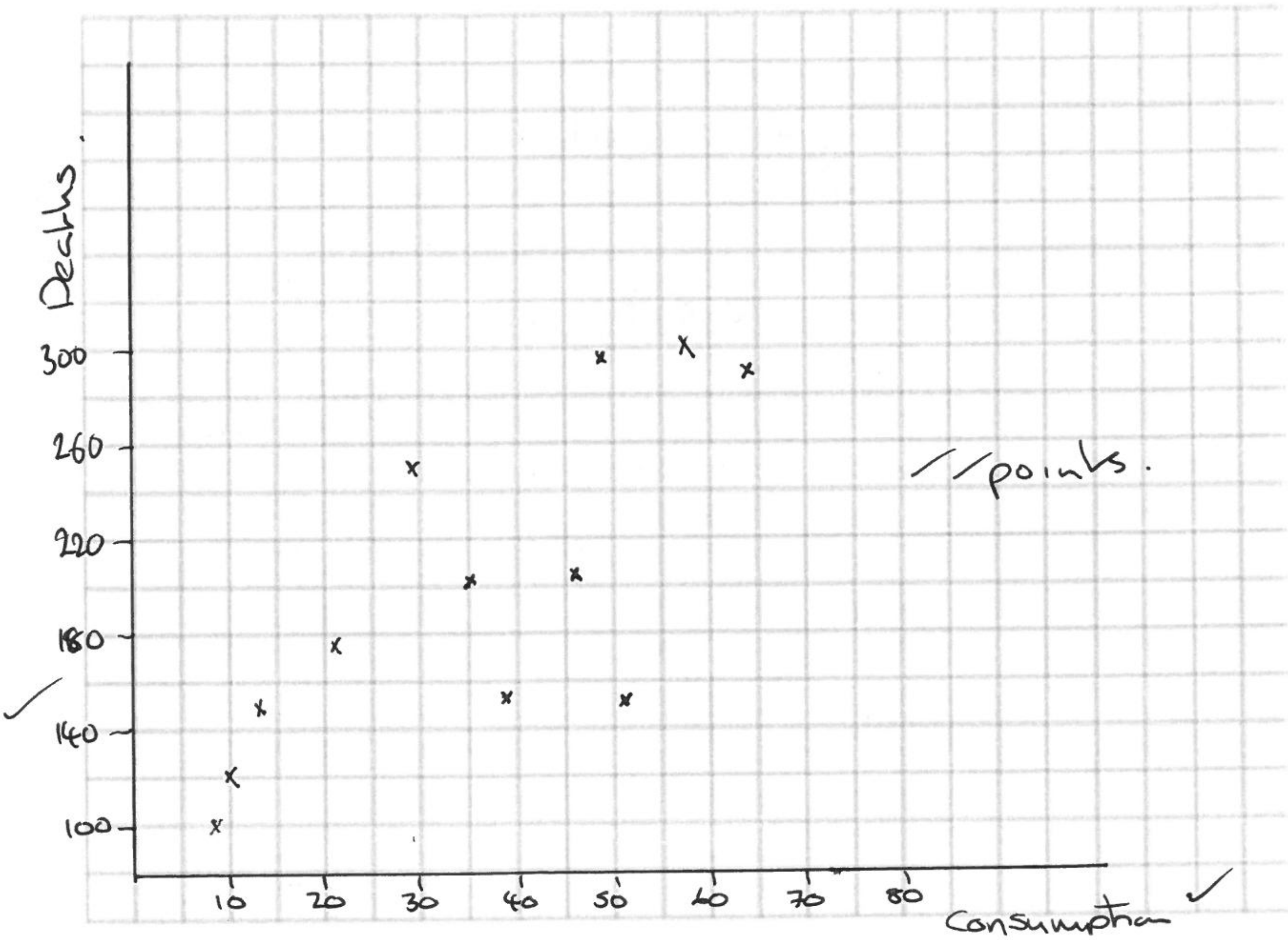
Answer using Line of best fit.

Q4) A statistician was asked to investigate whether countries with higher levels of consumption of a particular foodstuff tended to have a higher death rate due to heart disease. The following information was obtained for 12 countries.

[4+1+2=7 Marks]

	CONSUMPTION OF FOODSTUFF kg per person per year	DEATHS DUES TO HEART DISEASE Per year per 100 000 of population
Country A	57	299
Country B	13	154
Country C	51	152
Country D	48	295
Country E	35	208
Country F	9	100
Country G	21	175
Country H	10	120
Country I	46	202
Country J	64	290
Country K	38	156
Country L	29	250

a. Draw a scatterplot on the axis below.



b. Fit a trend line to the data

c. Krystal says that a country where people eat 50kg's of food would have 250,000 deaths in a year. Does this seem reasonable? Yes reason

Calculator Allowed Section – 20 minutes   Name ______________________

Q5) Consider the variables x and y and the corresponding bivariate data.

[2+2+1+2+1+1=9 Marks]

x	1	2	3	4	5	6	7	8
y	2	3	3	5	4	7	6	7

a   Construct a scatter plot for the data.

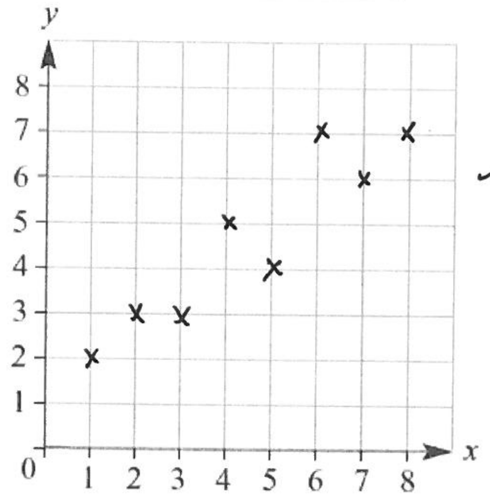
b   Describe the correlation between x and y?

linear positive strong

✓✓

c   Fit a line of best fit by eye to the data on the scatter plot.

✓



✓✓ points.

d   i   Use a graphics or CAS calculator or software to find the equation of the least squares regression line.

$y = 0.73x + 1.36$ ✓✓

ii   Estimate the value of y using the least squares regression line when $x = 9.5$.

≈ 8.3 ✓

iii   Estimate the value of x using the least squares regression line when $y = 7.8$.

≈ 12.5 ✓

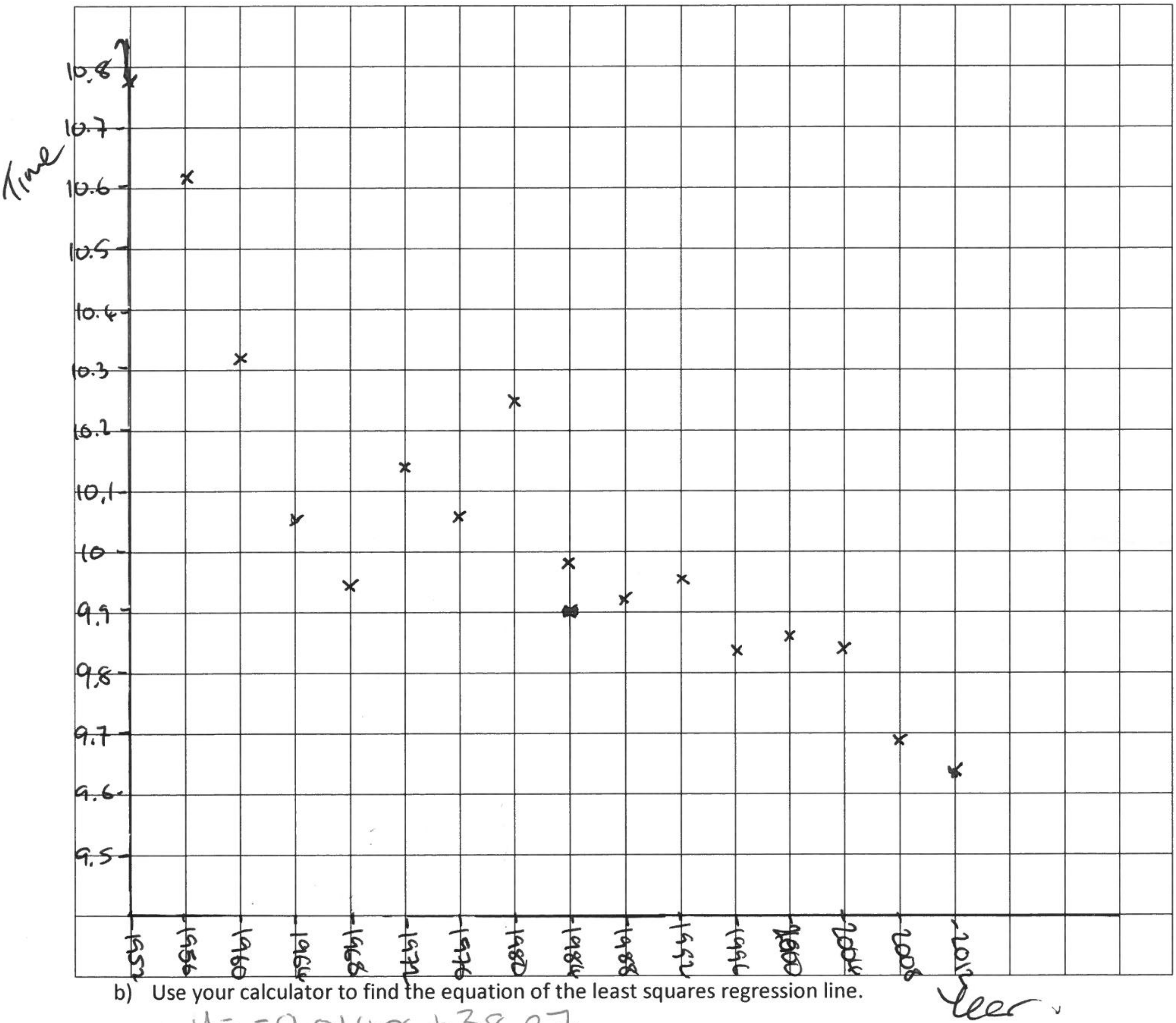
Q6) Usain Bolt of Jamaica successfully defended his title and set a new race record of 9.63 seconds in the 100m sprint in the London 2012 Olympic Games. Consider this record of winning times for the 100m sprint since 1952.

[4+2+2 = 8 marks]

Year	1952	1956	1960	1964	1968	1972	1976	1980
Time	10.79	10.62	10.32	10.06	9.95	10.14	10.06	10.25

Year	1984	1988	1992	1996	2000	2004	2008	2012
Time	9.99	9.92	9.96	9.84	9.87	9.85	9.69	9.63

a) Draw a scatterplot of the data.



b) Use your calculator to find the equation of the least squares regression line.

$y = -0.014x + 38.27$

c) In the 1960's, the speculation was 'When will the 10-second carrier be broken?' Today, talks centre around 'When will the 9.5 -second barrier be broken' Use your equation to make a prediction when this might occur.

$x = 2021.25$ So 2024 Olympics.

2021

~END OF TEST~